

MB 523: Introduction to AI and Automation

Course Code	MB 523
Course Name	Introduction to AI and Automation
L-T-P-C	2-0-0-2
Prerequisites	None
Intended for	MBA
Mutual Exclusion	None
Approval	52nd BoA (Nov 2023)

Course Contents

- Introduction to AI and Automation: Digitization and digitalization; Automation: traditional automation and intelligent automation; Identifying tasks and their characteristics suitable for automation, Automation payoffs; Defining AI, Types of AI: artificial narrow intelligence (ANI), artificial general intelligence (AGI) and artificial super intelligence (ASI), AI and machine learning, AI classification as per business capabilities- Process Automation, Cognitive Insights and Cognitive Engagement. (4 Lectures)
- Traditional Automation- No human Intervention: Scopes and techniques of automation, Business processes, Process Standardization, Reengineering and deconstruction, Identification of repetition, replication and redundancy elements in business process, Robotic Process Automation (RPA)- Scope, Benefits, Types and components of RPA, RPA Platforms, RPA vs. BPM, understanding RPA through use cases. (4 Lectures)
- AI for Automation: Automating decisions, human and machine capabilities for automation, Right Human-Automation Combination; Automating through Adaptive Business Processes, AI for automating repetitive tasks, AI for collaborative Automation (Human and Machine Collaboration). (6 Lectures)
- AI and Humans- Optimal Collaboration: Collaborative Automation through activities by humans and machines to complement each other's roles, human-in-the-loop systems, Humans for machine capability extension and amplification - Human trainers, explainers and sustainers for AI system development. AI augmenting human capabilities, Types of augmentation- Amplification, Interaction and Embodiment. (4 Lectures)
- Reimagining Business Processes: Proper mindset to reimagining processes, Experimentation, Leadership challenges for creating a blended culture of both humans and machines, and Building end-to-end data supply chains. (6 Lectures)
- New Skill Requirements for AI Based Automation: Skills required in the collaborative human and machine environment: Intelligent Interrogation, Bot Based Empowerment, Reciprocating Apprenticeship, Holistic Melding, Rehumanising Time, Responsible Normalizing, Judgement Integration, Relentless Reimagination. (4 Lectures)

References

1. Paul R. Daugherty and H. James Wilson, **Human+ Machine: Reimagining Work in the Age of AI**, Harvard Business Review Press, 2018.
2. Tom Taulli, **Artificial Intelligence Basics: A Non-technical Introduction**, Apress, 2019.
3. Thomas H. Davenport, **The AI Advantage: How to Put the Artificial Intelligence Revolution to Work**, The MIT Press, 2018
4. Ravin Jesuthasan and John W. Boudreau, **Reinventing jobs: a 4-step approach for applying automation to work**, Harvard Business Review Press, 2018.
5. Melanie Mitchell, **Artificial Intelligence: A Guide for Thinking Humans**, Pelican Books, Penguin Random House, 2019.
6. Tomas Chamorro-Premuzic, **I, Human: AI, Automation, and the Quest to Reclaim What Makes Us Unique**, Harvard Business Review Press, 2023.
7. Ian Barkin, Jochen Wirtz, and Pascal Bornet, **Intelligent Automation: Learn how to harness Artificial Intelligence to boost business & make our world more human**, Independently Published, 2020.
8. Ian Barkin, Jochen Wirtz, and Pascal Bornet, **Intelligent Automation: Learn how to harness Artificial Intelligence to boost business & make our world more human**, Independently Published, 2020.
9. Byron Reese, **The Fourth Age: Smart Robots, Conscious Computers, and the Future of Humanity**, Atria International, 2018.
10. Deepak Karwal, **The Automated Enterprise: Digital Reinvention through Intelligent Automation**, Deepak Karwal, 2020.
11. Matt Calkins, Neil Ward-Dutton, George Westerman et al., **Hyperautomation**, BookBaby, 2020.
12. Bhasker Ghosh, Gayathri Pallail and Rajendra Prasad, **The Automation Advantage: Embrace the Future of Productivity and Improve Speed, Quality, and Customer Experience Through AI**, McGraw-Hill Education, 2022.